

TABLE II
DIFFERENCE BETWEEN VARIOUS SOLAR PANELS

Property	Monocrystalline	Multi/polycrystalline	Thin-film
Efficiency	Highest	Moderate	Lowest
Cost	Highest	Moderate	Lowest
Area occupied per kW	Lowest	Moderate	Highest
High temperature performance	Poor	Poor	Better
Generation in diffused light	Average	Average	Better

Why select only the correct solar panel

You must select the right panel to:

- Ensure optimum energy is derived from the panels
- Avail the government's schemes
- Reduce electricity bills in the long run
- Reduce your carbon footprint
- Ensure long-term benefits

Brand. Good brands command a higher price, while cheap, imported Chinese panels cost less.

Testing standards. Panels that conform to government approvals and testing standards meet a minimum amount of performance and safety conditions; hence, are costlier.

Key pointers for selecting the right solar panel

Heat tolerance and temperature co-efficient of solar panels. This means that if a panel is rated as 100W, it should generate 100W under standard test conditions. In case the maker says that the panels have a tolerance of ± 5 per cent, it implies that the same panel can give either 105W or even 95W. Hence, you must ensure that your solar panel has a positive tolerance mentioned on the datasheet.

Temperature. Generally, panels are rated as per standard test conditions, that is, 25°C. If temperature is higher than this, panels may give less than the rated output.

Government and third-party certifications. These are as follows:

1. Panel-type certifications
2. Crystalline silicon terrestrial

PV modules: IEC 61215/ IS 14286

3. Thin-film terrestrial PV modules: IEC 61646

Also, panels must conform to IEC 61730 part I requirements for construction and part II requirements for testing and safety qualifications.

Warranty and guarantee. Indian manufacturers give one to five years warranty. Some also give linear performance guarantee, which is even better. A good panel should last as much as 25 years and perform faultlessly for at least ten years.

Installation costs. Although cost of a solar panel constitutes 70 per cent of the total cost of the solar power system, there is also an additional installation cost that must be kept in mind before doing a budget calculation. More the power required by your operations, more will be the number of panels to be installed. This will lead to higher cost.

Location and condition of the panel. Ensure that the panel is facing sunlight as even a single partially-shaded panel adversely impacts the output of all solar panels in the system. Keep the panels free of dust to avoid shading. The panels should be kept at an angle that will help in harvesting optimum energy from the sun.

Voltage. Most solar panels are designed to provide 16V, which can be used to charge a 12V battery.

Robust design. It must be made of tough material to withstand extreme climatic conditions.

Financial assistance. To comply with the government's net-metering guidelines, each household should ideally install solar panels capable of generating at least 1kW of solar energy. **EFY**

The buyers' guide titled Wireless Printers for Home announced for this issue earlier will be published later.

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